Lecture 9 — The 1D Heat Equation

18.085 Computational Science and Engineering I • MW 9:30–11, 6-120 • Wednesday, October 7, 2026

After this lecture you should be able to

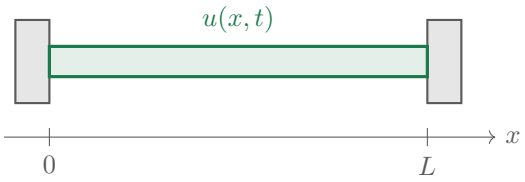
- Derive the continuity equation for heat in a rod by balancing energy on a short segment over a short time.
- State Fourier's law of cooling and explain the sign in $f = -k\,\partial u/\partial x$.
- Combine the two into the 1D heat equation and identify what the physical constants do and do not affect.
- Translate "fixed temperature" and "insulated" ends into boundary conditions on u and on $\partial u/\partial x$.

Where we are

Everything so far has been discrete: grids, difference matrices, eigenvalues, pivots. Today we go the other way and build, from physics, the continuous problem that those matrices have been approximating. The pattern of the derivation matters as much as the answer: a *conservation law* that is exact bookkeeping, plus a *constitutive law* that describes the material, combine into one second-order equation.

1 The physical setup

Denote the temperature of the rod by $u(x,t)$, which is a function of both the position x and the time t.



Let the linear density of the material in the rod be σ, and the heat capacity per mass be C. Both are constants of the material.

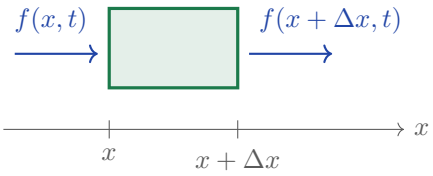
The temperature distribution in the rod changes because energy in the form of heat is flowing from high temperatures to low temperatures. Denote the heat flux in the rod by $f(x,t)$, and specify that f is positive if heat is flowing from left to right, and negative vice versa.

Two functions, one unknown

We have introduced two unknown fields, u and f, on the same rod. The derivation below produces one equation relating them (conservation of energy) and then one more equation expressing f in terms of u (Fourier's law). Two equations, two unknowns — and eliminating f leaves a single equation for u alone.

2 Conservation of energy on a short segment

To derive the heat equation, consider a very short segment of the rod, x to $x + \Delta x$, during a very short interval of time, t to $t + \Delta t$.



Since the segment is very short, we can approximately view its temperature as being uniform: $u(x,t)$ at time t, and $u(x,t+\Delta t)$ at time $t+\Delta t$. Write Δu for the difference.

What causes u to change? The net heat flow into the segment over Δt. Heat enters at the left at rate $f(x,t)$ and leaves at the right at rate $f(x+\Delta x,t)$, so

$$\text{Net heat flow} = \big(-f(x+\Delta x,t) + f(x,t)\big)\cdot \Delta t.$$

That same quantity of energy can be counted a second way — through what it does to the temperature:

$$\text{Net heat flow} \;=\; \Delta u \cdot \underbrace{C \cdot \overbrace{\sigma \Delta x}^{\text{mass of segment}}}_{\text{heat capacity of segment}},$$

the net heat required to change the rod's temperature by Δu. Setting the two expressions equal and dividing by $C\sigma\,\Delta x\,\Delta t$,

$$\frac{\Delta u}{\Delta t} = -\frac{1}{C\sigma}\,\frac{f(x+\Delta x,t) - f(x,t)}{\Delta x},$$

and taking the limits $\Delta t \to 0$, $\Delta x \to 0$:

$$\frac{\partial u}{\partial t} = -\frac{1}{C\sigma}\,\frac{\partial f}{\partial x}. \tag{1}$$

The continuity equation

Equation (1) is known as a **continuity equation**, because it is a statement that heat is not created nor destroyed: it just flows from one place to the other. The temperature at a point can only rise if more heat arrives than leaves, and $-\partial f/\partial x$ is precisely that imbalance.

Notice that nothing in (1) is a modelling assumption about heat. It is exact bookkeeping, and it would look the same for mass, charge, or probability. What is missing is any statement about *why* heat flows.

3 Fourier's law of cooling

Now we just need to express the heat flow f in terms of the temperature distribution u:

$$f = -k\,\frac{\partial u}{\partial x}. \tag{2}$$

Properly, laws like this should be derived from a microscopic theory of heat; for the purposes of this class we will take this as a given.

Intuitively, Fourier's law of cooling states that heat flows from hot to cold at a rate proportional to the temperature gradient. The minus sign is the whole physical content: if the temperature increases to the right ($\partial u/\partial x > 0$), then heat flows to the left, so $f < 0$. The constant k is the conductivity of the material.

Conservation law vs. constitutive law

Equation (1) is a conservation law: exact, material-independent, non-negotiable. Equation (2) is a constitutive law: it describes *this* material, it is approximate, and it is where all the physics of the substance lives. Almost every continuum model in this course is assembled from exactly this pair. Keeping the two kinds of statement separate is a habit worth forming now.

4 Combining the two

Substituting (2) into (1) gives, for constant k,

$$\frac{\partial u}{\partial t} = -\frac{1}{C\sigma}\frac{\partial}{\partial x}\left(-k\frac{\partial u}{\partial x}\right) = \boxed{\frac{k}{C\sigma}\frac{\partial^2 u}{\partial x^2}} \qquad \text{(1D heat equation)}.$$

The three physical constants enter only through the single *global factor* $k/(C\sigma)$ in front of the second derivative. In class we just took the global factor to be 1, so that the equation reads

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}.$$

Key idea

The 1D heat equation says: the rate at which a point heats up is proportional to the *curvature* of the temperature profile there. A point sitting in a dip ($\partial^2 u/\partial x^2 > 0$) is colder than its neighbours on average, so it warms; a point at a bump cools. Diffusion is the continuous statement that everything is pulled towards the average of its neighbours — which is exactly what the $1, -2, 1$ stencil computes.

A preview of the future

With more spatial dimensions, the flux becomes a vector field $\boldsymbol{f}$ and the two laws generalise directly. Equation (1) becomes

$$\frac{\partial u}{\partial t} = -\frac{1}{C\rho}\nabla \cdot \boldsymbol{f},$$

and (2) becomes $\boldsymbol{f} = -k\,\nabla u$, so the heat equation reads

$$\frac{\partial u}{\partial t} = \frac{k}{C\rho}\nabla^2 u.$$

(The linear density σ has become a volume density ρ.) The gradient ∇ measures the temperature difference, the material constant k turns it into a flux, and the divergence $\nabla\cdot$ balances what flows in against what flows out. That three-step chain — difference, material constant, balance — is the same $\boldsymbol{A}$, $\boldsymbol{C}$, $\boldsymbol{A}^{\mathsf{T}}$ structure we will build matrices for in Lecture 12.

5 Boundary conditions

The heat equation on its own does not determine u: we must also say what happens at the two ends of the rod. In the following we will only consider the boundary at $x = 0$ for convenience. Boundary conditions at other locations are analogous.

① A fixed temperature boundary

$$u(x = 0, t) = 0.$$

The rod is free to exchange heat with a thermal mass of fixed temperature at $x = 0$, so the temperature of the rod at the boundary remains constant. The reservoir supplies or absorbs whatever heat is needed to hold that value.

② An insulated boundary

$$\frac{\partial u}{\partial x}(x = 0, t) = 0.$$

The rod is insulated at the $x = 0$ end: no heat flux is allowed through. Per Fourier's law of cooling, $f = -k\,\partial u/\partial x$, so demanding $f = 0$ at the boundary is exactly the same as demanding that the temperature gradient at $x = 0$ must be 0.

Key idea

The condition on the *value* u and the condition on the *slope* $\partial u/\partial x$ are the continuous originals of the two boundary conditions we imposed on the grid back in Lecture 2 — fixed and free. Fixed temperature gives the fixed end of $\boldsymbol{K}$; insulated gives the free end of $\boldsymbol{T}$. The boundary condition is not a technicality attached to the equation; it is a second piece of physics.

An insulated end is not a cold end

"No heat flows through the boundary" does not mean "the boundary stays at zero temperature." It is the opposite: heat piles up against an insulated wall. The two conditions constrain different objects — one the value of u, the other its x-derivative — and mixing them up is the most common way to solve the wrong physical problem correctly.

Where this is going

Next time we return to linear algebra with the QR decomposition and orthogonality, which is the machinery behind least squares. The heat equation comes back when we discretise it: $\partial^2 u/\partial x^2$ becomes the $1, -2, 1$ matrix, the two boundary conditions above select $\boldsymbol{K}$ or $\boldsymbol{T}$, and the positive definiteness we established in Lecture 8 is what forces the temperature to relax rather than blow up.

Reading

Strang, *Computational Science and Engineering*, § 6.5 (the diffusion / heat equation); § 1.2 for the fixed and free boundary conditions in their discrete form.